Synoptic Question 8: Photovoltaic Solar Power Generation

Reading Passage

Read the following article and then answer the questions that follow.

Photovoltaic (PV) cells convert sunlight directly into electricity using the photoelectric effect in a semiconductor material, typically silicon. When a photon of sunlight with sufficient energy strikes the silicon, it can dislodge an electron from its atom, creating an electron-hole pair. An internal electric field within the cell separates these charges, causing the electrons to flow to one side and the holes to the other. This separation creates a potential difference (voltage). When an external circuit is connected, this flow of electrons forms a direct current (DC).

The total power output of the Sun is immense, approximately 3.8×10^{26} W. This energy radiates outwards in all directions. The intensity of this radiation decreases with the square of the distance from the Sun. At the top of Earth's atmosphere, this intensity, known as the "solar constant", is about 1360 W m^{-2}.

A key performance indicator for a PV panel is its Current-Voltage (I-V) characteristic curve, which shows the relationship between the current it produces and the voltage across it under specific conditions of sunlight intensity (irradiance) and temperature. The power output ($P = VI$) is not constant; there is a unique "maximum power point" (P_{mp}) on the curve where the product of voltage (V_{mp}) and current (I_{mp}) is highest.

A household in a sunny region installs a rooftop solar array to reduce its electricity bills. Each panel in the array has its performance characterized by the graph in Figure 8.1.

Data and Graph:

Table 1: Data for Solar System and Physics

Parameter	**Value**
Total power output (Luminosity) of the Sun, L_{Sun}	3.8×10^{26} W
Mean distance from Earth to Sun, R	1.5×10^{11} m
Single solar panel area, A_{panel}	1.65 m^2
Mass of a single panel, m_{panel}	18.0 kg
Specific heat capacity of panel, c_{panel}	850 J kg^{-1} K^{-1}
Speed of light, c	3.00×10^8 m s^{-1}
Planck's constant, h	6.63×10^{-34} J s
Band gap of silicon, E_g	1.12 eV
Charge of one electron, e	1.60×10^{-19} C

Questions

(a) Using the Sun's luminosity and its distance from Earth, calculate the solar intensity (solar constant) at the top of Earth's atmosphere. (Assume the Sun's power is distributed over the surface of a sphere of radius R).

(b) The graph in Figure 8.1 is measured at an irradiance of 1000 W m^{-2}. Give two reasons why this is less than the value calculated in (a).

(c) From the graph (Figure 8.1), determine:

  (i) The open-circuit voltage, V_{oc} (the voltage when the current is zero).

  (ii) The short-circuit current, I_{sc} (the current when the voltage is zero).

(d) The power output of the panel is the product of voltage and current ($P = VI$).

  (i) Estimate the voltage (V_{mp}) and current (I_{mp}) at the maximum power point, P_{mp}, which is marked on the graph.

  (ii) Calculate the maximum power, P_{mp}, for this panel.

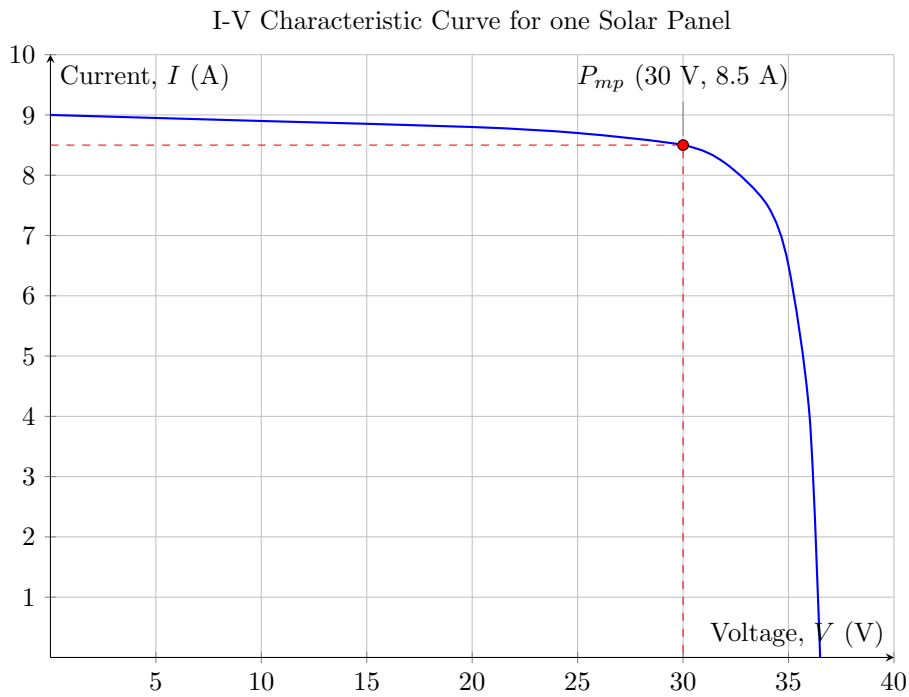

Figure 1: I-V Curve for one panel at 25 °C and an irradiance of 1000 W m^{-2}.

(iii) The "fill factor" of a panel is a measure of its quality, given by $FF = \frac{P_{mp}}{V_{oc} \times I_{sc}}$. Calculate the fill factor for this panel.

(e) Using the maximum power from (d)(ii) and the data provided, calculate the efficiency (η) of the solar panel under the conditions of the test.

(f) The household array consists of 20 of these panels connected together. If the array produces an average of 35 kWh of energy per day, calculate the total monetary savings for the household over one year (365 days). The cost of electricity from the grid is \$0.28 per kWh.

(g) The operation of the PV cell relies on the photoelectric effect.

(i) Convert the band gap energy of silicon, 1.12 eV, into joules.

(ii) Calculate the maximum wavelength of a photon that has just enough energy to create an electron-hole pair in silicon.

(iii) What happens to photons with a wavelength longer than this maximum value?

(h) On a hot day, the panel (mass 18.0 kg) absorbs solar energy and its temperature increases from 25.0 °C to 55.0 °C. Calculate the total thermal energy absorbed by the panel.

Solutions

(a) Intensity $I = \frac{\text{Power}}{\text{Area}} = \frac{L_{Sun}}{4\pi R^2}$
$I = \frac{3.8\times10^{26}}{4\pi(1.5\times10^{11})^2} = 1344$ W m^{-2}.
(This is consistent with the known solar constant of approx. 1360 W m^{-2}).

(b) Two reasons for lower intensity at the surface:

- **Absorption:** Gases in the atmosphere (like ozone, water vapour) absorb some of the radiation.
- **Reflection/Scattering:** Clouds and dust particles in the atmosphere reflect and scatter sunlight back into space.

(c) From the graph:

(i) V_{oc} (x-intercept) ≈ 36.5 V.

(ii) I_{sc} (y-intercept) ≈ 9.0 A.

(d) (i) From the marked point P_{mp} on the graph:
$V_{mp} \approx 30.0$ V.
$I_{mp} \approx 8.5$ A.

(ii) $P_{mp} = V_{mp} \times I_{mp} = (30.0) \times (8.5) = 255$ W.

(iii) $FF = \frac{P_{mp}}{V_{oc}\times I_{sc}} = \frac{255}{(36.5)\times(9.0)} = \frac{255}{328.5} = 0.776$.

(e) Efficiency $\eta = \frac{\text{Power Out}}{\text{Power In}} = \frac{P_{mp}}{I_{incident}\times A_{panel}}$
The test conditions are 1000 W m^{-2} irradiance.
Power In = (1000 W m^{-2}) $\times$ (1.65 m^2) = 1650 W.
$\eta = \frac{255 \text{ W}}{1650 \text{ W}} = 0.1545$.
Efficiency $\approx 15.5\%$.

(f) Total energy per year = (35 kWh/day) $\times$ (365 days/year) = 12775 kWh.
Total savings = Total energy $\times$ Cost per kWh
Total savings = $12775 \times 0.28 = \$3577$.

(g) (i) $E = (1.12 \text{ eV}) \times (1.60 \times 10^{-19} \text{ J/eV}) = 1.792 \times 10^{-19}$ J.

(ii) $E = hf = \frac{hc}{\lambda_{max}}$.
$\lambda_{max} = \frac{hc}{E} = \frac{(6.63\times10^{-34})(3.00\times10^8)}{1.792\times10^{-19}} = 1.11 \times 10^{-6}$ m (or 1110 nm, which is in the infrared spectrum).

(iii) Photons with a longer wavelength (e.g., infrared light) have less energy than the 1.12 eV band gap. They cannot dislodge an electron and will pass through the silicon or be absorbed as heat, without generating any electric current.

(h) $\Delta T = 55.0$ °C $-$ 25.0 °C = 30.0 °C (or 30.0 K).
Thermal Energy $Q = mc\Delta T$
$Q = (18.0 \text{ kg}) \times (850 \text{ J kg}^{-1}\text{K}^{-1}) \times (30.0 \text{ K})$
$Q = 459000$ J (or 459 kJ).

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Variant 2 (Higher Difficulty)

Topic Focus: Photovoltaic Systems and Applied Physics Concepts

Instructions (CIE P4 style)

- Answer all questions.
- Show all working clearly and include units.
- Use a value of $g = 10\,\mathrm{m\,s^{-2}}$ unless otherwise stated.
- Calculators may be used.

Question 1: Solar irradiance, panel efficiency, and uncertainty [12]

A photovoltaic (PV) installation uses 24 identical panels. Each panel has area $1.60\,\mathrm{m}^2$. At midday, the incident solar irradiance is measured as $920\,\mathrm{W\,m^{-2}}$, and total electrical output power is $5.30\,\mathrm{kW}$.

(a) Calculate the total incident solar power on the array. [2]

(b) Determine the conversion efficiency of the array. [2]

(c) The irradiance meter has an uncertainty of $\pm 3\,\%$ and output power has uncertainty $\pm 2\,\%$. Estimate the percentage uncertainty in efficiency. [3]

(d) State two physical reasons why the measured efficiency is lower than the quoted cell material efficiency under laboratory conditions. [2]

(e) Open-ended: A school claims that "installing a mirror reflector behind each panel will always increase annual energy output." Discuss this claim using physics arguments, giving at least three relevant factors and a justified conclusion. [3]

Question 2: Semiconductor physics in photovoltaic cells [11]

A silicon solar cell forms a p–n junction.

(a) Explain how doping creates p-type and n-type semiconductors. [2]

(b) Describe how the depletion region and built-in electric field are formed at the junction. [3]

(c) Explain how incident photons produce a photocurrent in an external circuit. [3]

(d) Open-ended: The band gap of a PV material is changed from $1.1\,\mathrm{eV}$ to $1.8\,\mathrm{eV}$. Discuss how this may affect output voltage, current, and overall power in sunlight. [3]

Question 3: I–V characteristics and load matching [13]

A PV panel has an open-circuit voltage $V_{oc} = 22.0\,\mathrm{V}$ and short-circuit current $I_{sc} = 8.4\,\mathrm{A}$. At maximum power point (MPP), $V_{mpp} = 17.5\,\mathrm{V}$ and $I_{mpp} = 7.6\,\mathrm{A}$.

(a) Calculate the maximum electrical power output. [2]

(b) Calculate the fill factor, $\mathrm{FF} = \dfrac{V_{mpp} I_{mpp}}{V_{oc} I_{sc}}$. [2]

(c) The panel is connected directly to a $3.0\,\Omega$ resistor. Estimate the operating current and power by assuming operation near $V \approx 18\,\mathrm{V}$ at that load point. State one limitation of this estimate. [4]

(d) Open-ended: Explain why maximum power point tracking (MPPT) improves energy yield over a fixed-resistance load during a day with changing cloud cover and temperature. [5]

Question 4: Thermal effects and energy transfer [12]

A panel of area $2.0\,\mathrm{m}^2$ absorbs $1400\,\mathrm{W}$ of solar power and delivers $310\,\mathrm{W}$ electrically. The rest is mainly transferred to the surroundings as thermal energy.

(a) Calculate the rate of thermal energy transfer from the panel. [1]

(b) The panel has mass $18\,\mathrm{kg}$, average specific heat capacity $900\,\mathrm{J\,kg^{-1}\,K^{-1}}$, and initially no heat loss. Estimate the initial rate of temperature rise. [3]

(c) In practice, heat loss occurs by radiation, convection and conduction. Explain how wind speed affects panel temperature and electrical efficiency. [3]

(d) Open-ended: Suggest and evaluate two engineering strategies to keep panel operating temperature lower in a hot climate, considering both physics effectiveness and practical trade-offs. [5]

Question 5: Storage, transmission and system design [12]

A remote clinic requires $18.0\,\mathrm{kWh}$ electrical energy per day. A battery bank has round-trip efficiency 85%, and inverter efficiency is 92%. Average daily energy produced by the PV array before storage/conversion losses is $26.0\,\mathrm{kWh}$.

(a) Calculate the daily usable energy delivered after battery and inverter losses. [3]

(b) Determine whether the system meets demand with this average production. [1]

(c) The site wants two days of autonomy (no solar input), with maximum depth of discharge 60%. Calculate the minimum battery energy capacity required (in kWh, rated capacity). [4]

(d) Open-ended: Discuss the physics and engineering reasons for using high transmission voltage from array to charge controller when cable runs are long. [4]

Sample Solutions – Variant 2

Q1

(a) Total area $A = 24 \times 1.60 = 38.4\,\mathrm{m}^2$. Incident power:

$$P_{in} = IA = 920 \times 38.4 = 35\,328\,\mathrm{W} = 35.3\,\mathrm{kW}.$$

(b) Efficiency:

$$\eta = \frac{P_{out}}{P_{in}} = \frac{5.30}{35.328} = 0.150 \approx 15.0\,\%.$$

(c) For division, add percentage uncertainties:

$$\frac{\Delta\eta}{\eta} \approx 3\% + 2\% = 5\%.$$

So efficiency uncertainty $\approx \pm 5\%$ (relative), i.e. absolute $\approx \pm 0.75$ percentage points.

(d) Any two: higher cell temperature; reflection losses; spectral mismatch; resistive losses; dust/shading.

(e) Valid discussion: reflector can increase irradiance but may raise panel temperature (reducing efficiency), create non-uniform illumination/hot spots, increase dirt accumulation/maintenance, seasonal sun-angle dependence, and possible glare/safety issues. Conclusion: not "always" beneficial; depends on thermal design and geometry.

Q2

(a) p-type: trivalent dopant creates holes as majority carriers. n-type: pentavalent dopant contributes extra electrons.

(b) Diffusion of electrons/holes across junction leaves fixed ions, forming depletion region and internal electric field from n to p side.

(c) Photon with energy $\geq E_g$ creates electron-hole pair; field separates charges, driving electrons toward n side and holes to p side, producing emf/current through load.

(d) Larger E_g: tends to increase per-carrier voltage but fewer photons exceed threshold, so photocurrent drops. Overall power may rise or fall depending on spectrum and material quality; very large E_g often lowers total power under broad sunlight.

Q3

(a) $P_{mpp} = 17.5 \times 7.6 = 133\,\mathrm{W}$ (133.0 W).

(b)

$$\mathrm{FF} = \frac{133.0}{22.0 \times 8.4} = 0.720 \approx 0.72.$$

(c) With $R = 3.0\,\Omega$, if $V \approx 18\,\mathrm{V}$:

$$I \approx \frac{V}{R} = \frac{18}{3.0} = 6.0\,\mathrm{A}, \quad P \approx VI = 18 \times 6.0 = 108\,\mathrm{W}.$$

Limitation: actual operating point depends on full non-linear I–V curve; assumed voltage may be inaccurate.

(d) MPPT continuously adjusts effective load so operating point remains near MPP despite irradiance/temperature changes; fixed resistance only intersects curve at one load line, usually away from MPP. As clouds and temperature alter V_{oc}, I_{sc}, and curve shape, MPPT captures more daily energy.

Q4

(a) $\dot{Q}_{thermal} = 1400 - 310 = 1090\,\mathrm{W}$.

(b) Initial heating rate:

$$\frac{\mathrm{d}T}{\mathrm{d}t} = \frac{\dot{Q}}{mc} = \frac{1090}{18 \times 900} = 6.73 \times 10^{-2}\,\mathrm{K\,s^{-1}} \approx 0.067\,\mathrm{K\,s^{-1}}.$$

(c) Higher wind speed increases convective heat transfer coefficient, lowering panel temperature; lower temperature usually increases voltage and therefore electrical efficiency/power.

(d) Example strategies: (i) rear ventilation gap / heat-sink fins to improve convection; (ii) high-emissivity backsheet for radiative cooling; (iii) water cooling (effective but complexity/water use); (iv) spacing/orientation to reduce heat build-up. Evaluation should include cost, reliability, maintenance, and local climate suitability.

Q5

(a) Usable energy:

$$E_{use} = 26.0 \times 0.85 \times 0.92 = 20.3\,\mathrm{kWh\,day^{-1}} \text{ (3 s.f.)}.$$

(b) Since $20.3 > 18.0$, average demand is met (surplus $\approx 2.3\,\mathrm{kWh\,day^{-1}}$).

(c) Energy needed at load for 2 days: $2 \times 18.0 = 36.0\,\mathrm{kWh}$. Battery must supply pre-inverter energy: $36.0/0.92 = 39.1\,\mathrm{kWh}$. At max DoD 60%, rated capacity

$$C_{rated} = \frac{39.1}{0.60} = 65.2\,\mathrm{kWh}.$$

(d) High voltage lowers current for same power ($P = VI$); cable loss $P_{loss} = I^2R$ decreases strongly, allowing smaller cables and reduced voltage drop. Trade-offs: insulation/safety requirements and converter complexity.

Mark Scheme – CIE P4 Style (Variant 2 Only)

Total: 60 marks

Question 1 [12]

- (a) Correct total area (1), correct incident power with unit (1)
- (b) Correct efficiency method (1), correct value $\approx 15\%$ (1)
- (c) Adds relative uncertainties for quotient (1), gets 5% (1), interprets as efficiency uncertainty (1)
- (d) Any two valid physics reasons (1+1)
- (e) Open-ended: relevant factors (max 2), justified conclusion (1)

Question 2 [11]

- (a) Correct p-type statement (1), correct n-type statement (1)
- (b) Carrier diffusion/recombination mention (1), depletion region (1), built-in field direction/purpose (1)
- (c) Photon energy threshold and e-h pair creation (1), separation by field (1), external current explanation (1)
- (d) Open-ended: effect on voltage (1), effect on current (1), balanced power conclusion (1)

Question 3 [13]

- (a) Correct power expression and value (2)
- (b) Correct FF substitution and value (2)
- (c) Uses $I = V/R$ with stated assumption (1), correct current (1), correct power (1), valid limitation (1)
- (d) Open-ended MPPT discussion: curve/load-line idea (2), changing weather/temperature effects (2), clear conclusion (1)

Question 4 [12]

- (a) Correct thermal power difference (1)
- (b) Uses $\mathrm{d}T/\mathrm{d}t = \dot{Q}/(mc)$ (1), substitution (1), correct value with unit (1)
- (c) Wind–convection link (1), effect on temperature (1), link to efficiency/power (1)
- (d) Open-ended strategies: two valid methods (2), physics evaluation of effectiveness (2), practical trade-off judgement (1)

Question 5 [12]

- (a) Sequential efficiency multiplication method (1), correct value (2)
- (b) Correct comparison with demand and statement (1)
- (c) 2-day demand energy (1), inverter adjustment (1), DoD adjustment method (1), final capacity value (1)

- (d) Open-ended transmission discussion: low-current/I^2R logic (2), voltage drop/cable sizing (1), safety/engineering trade-off (1)

Examiner note: For open-ended items, award marks for scientifically correct reasoning, clear linkage to core physics principles, and a justified conclusion even where wording differs from sample solution.